

Lowpass-Notch and Highpass-Notch Filters

Get to know these lopsided Cinderellas of the notch family

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The lowpass notch and highpass notch are poorly understood but important members of the family of second order filters and are important building blocks of elliptical filters. When an elliptical filter response is written in terms of first and second order stages the second order stages consist exclusively of highpass or lowpass notches. It is very useful to understand how these notch filters behave in order to be able to realize and implement an elliptical filter (or similar type of filter) via analog circuitry or in digital form.

The continuous-time transfer function for the notch filter has the following form:

$$H(s) = H_0 \cdot \frac{s^2 + \omega_z^2}{s^2 + \frac{\omega_p}{Q} s + \omega_p^2} \quad \text{Equation 1}$$

The symbol ω refers to the frequency in radians, where $\omega = 2\pi F$ and F is the frequency in Hertz. The pole and zero frequencies, ω_p and ω_z , may take on dissimilar values. When $\omega_p < \omega_z$ the response shape is termed a lowpass notch. When $\omega_p > \omega_z$, the response shape is termed a highpass notch. The symmetric notch is obtained if $\omega_p = \omega_z$. Figure 1 provides examples of the various response shapes that equation 1 can produce. For the symmetric notch function both the pole and zero frequencies are 1.0k Hz. For the lowpass notch response the pole location is at approximately 0.6k Hz and the zero location at approximately 2.3k Hz. For the highpass notch the pole location is at 2.5k Hz and the zero location at approximately 0.31k Hz. The Q value is 3 in each case. When the pole and zero frequencies are sufficiently far apart the response in between them has a constant slope of 12dB per octave. Although the lowpass and highpass notch responses in some ways resemble a second order shelf filter plus a symmetric notch, these features all arise from a single second order function.

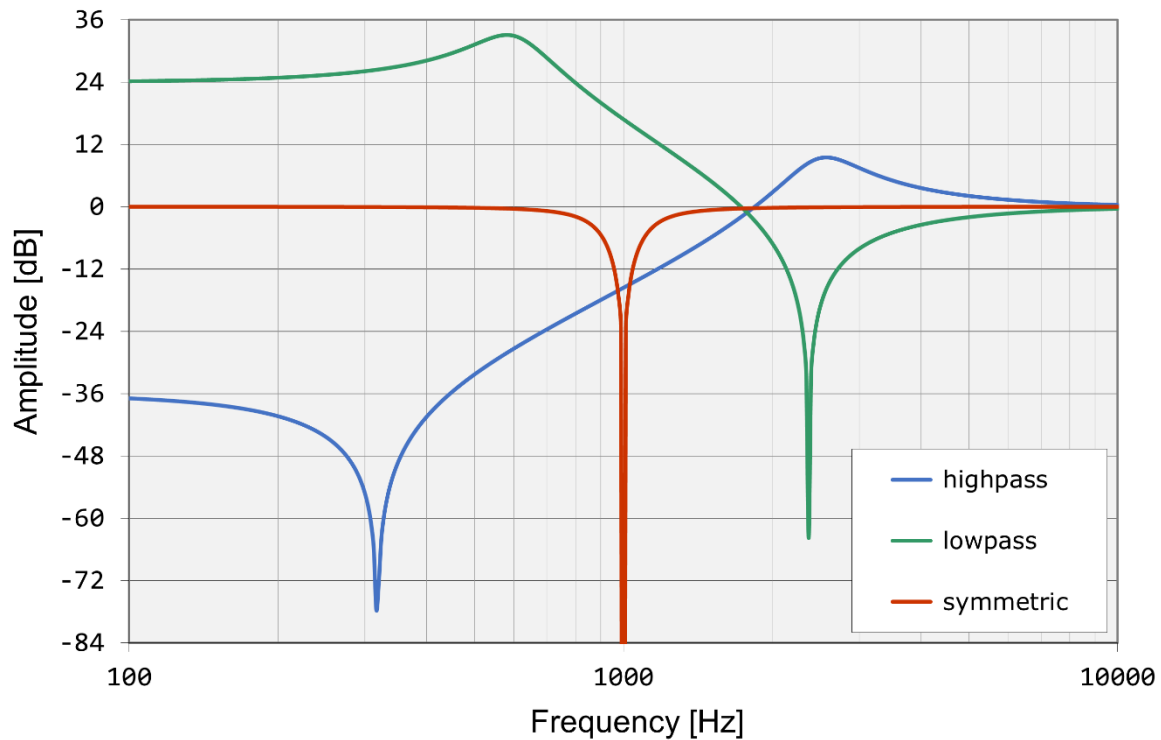


Figure 1: The three types of notch responses.

There are a couple of peculiarities to the highpass and lowpass notch responses. Both responses have a gain of unity at high frequency and only their low frequency responses are different. This has a strange consequence for the passband of the lowpass notch response. Typically, when considering a filter response, the passband region is where the signal is to remain unaltered such that a gain of 0dB is the expected level. This is clearly the case for the highpass notch response since the region above the pole frequency takes on the value of 0dB. In contrast, the passband region of the lowpass notch is found below the pole frequency but its stopband has a constant gain of 0dB while the passband gain is elevated. This odd behavior is simply a consequence of the notch filter transfer function. By inspecting the transfer function, we see that for very high frequencies (as $s \rightarrow \infty$) the gain will be 1 and at very low frequencies (as $s \rightarrow 0$) the gain will be $(\omega_z/\omega_p)^2$. When expressed in Decibels, the low frequency gain of the notch filter is $40 \log_{10}(\omega_z/\omega_p)$.

The DC gain of the lowpass notch filter creates problems when you attempt to implement the transfer function given by eq. 1 directly. In order to have 0dB passband gain, the entire lowpass notch response must have its gain reduced by $40 \log_{10}(\omega_z/\omega_p)$, either by attenuating the input, attenuating the output, or using internal attenuation of some means. Because the ratio ω_z/ω_p can be as high as 10 or more, the required attenuation can exceed 40dB. When implemented with a real-world circuit this leads to serious degradation of the signal to noise ratio. No similar problem exists for the highpass notch circuit, however, so the challenge centers on how to practically implement the lowpass notch response.

One way to avoid excessive gain at low frequency is to create the entire Nth order lowpass notch via a passive network such as an RLC ladder filter because the passive network cannot generate gain over a wide band or at DC. A passive filter could operate at line level into a known resistive load and possibly buffered by active devices at the input and output. Unfortunately, designing a passive ladder filter, especially a higher order one, is not always straightforward and is beyond the scope of this article. Additionally, higher order passive RLC circuits tend to have complicated interactions among components such that tuning the response to closely match the filter specifications can be extremely difficult. This is a consequence of using real world components that have deviations from the desired values. Exact values may be difficult to achieve, especially with capacitors, and may drift. When corner frequencies are low, component values may become quite high and require large, bulky and expensive parts. Although successful passive LC ladder realization may be *possible*, let us set aside the passive network as a practical way to realize a lowpass or highpass notch filter.

Analog active filters may also be used to implement lowpass and highpass notch responses. There are several different and well-known notch filters but only some can implement the lowpass or highpass forms directly. The circuit by Bainter can do this, as well as the circuits by Bocktor, Scultety, and Friend's SAB [1]. Unfortunately, all of these suffer from the gain problem at low frequency, or other gain and noise issues in their output that would require attenuation of the signal and/or loss of fidelity. Another means of notch response synthesis is from a weighted sum of non-inverting second order lowpass and highpass filters that have identical pole frequency and Q, and notch responses formed with this approach do not require DC gain correction. The relative phase angle between these two filters is 180 degrees at all frequencies and their responses will cancel when summed at equal levels, forming a notch. Sallen-Key filters could be used to implement the required HP and LP filters. Because a weighted sum must be obtained, an inverting summing amplifier is required. To restore normal phase, a final unity gain inverting amplifier could be added or the inverting multiple-feedback topology could be used to implement the filters. This requires either three or four amplifiers and each of the second order filters requires two or three capacitors. The filters must be carefully tuned to have identical F and Q values in order to achieve a deep notch. There is, however, a better circuit that will do all of this with four amplifiers and only two capacitors - the state-variable filter shown in figure 2.

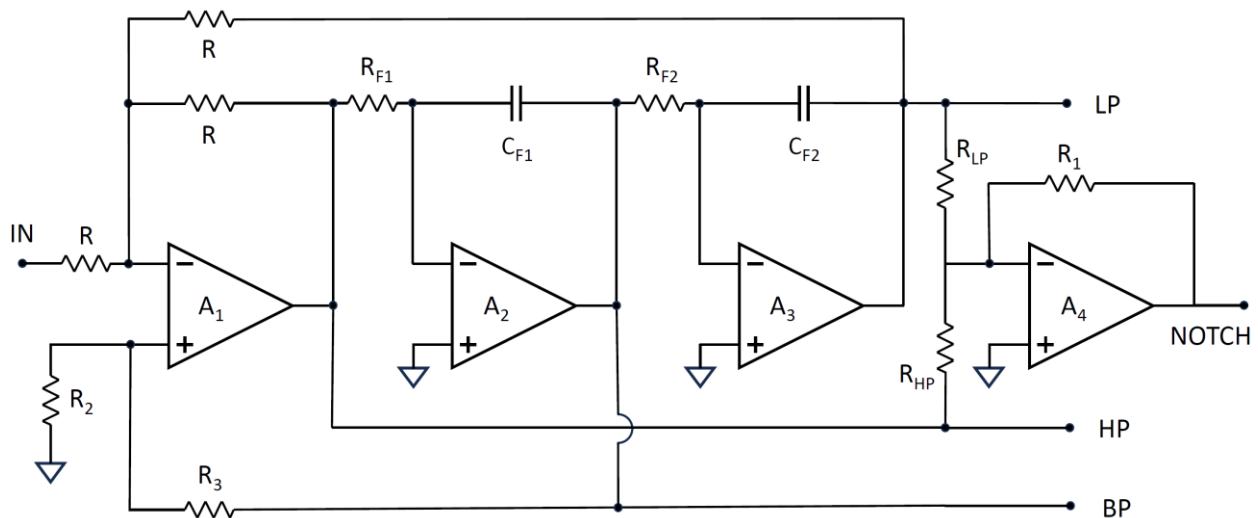


Figure 2: The state-variable filter with additional summing amplifier A4.

This three op-amp circuit was proposed by Kerwin, Huelsman, and Newcomb in 1967 [2]. It can simultaneously produce the lowpass response (LP), the highpass response (HP), a bandpass response (BP), and using an additional summing amplifier all three notch responses (NOTCH). One useful characteristic of this circuit is that the F and Q values of the HP, BP, and LP outputs will be identical because these responses are formed via the same two integrators (formed by amplifiers A2 and A3). The responses have low component sensitivities and the circuit allows for independent adjustment of F and Q. Because the input to the circuit is to the inverting input of amplifier A1 the lowpass and highpass filter outputs have inverted phase, however, inverting summing amplifier A4 restores normal polarity to the notch output by applying a gain of $-R_{\text{FEEDBACK}}/R_{\text{INPUT}}$ to each of its inputs.

Choosing component values for this circuit is very straightforward. First, determine an approximate value for the capacitors using the relationship $C = 4 \times 10^{-7} / F_P^{1/2}$. Round this value up or down to the next standard value. The required resistance values R_{F1} and R_{F2} used in the integrators formed by A2 and A3 are then calculated as $R = 1/(2\pi F_P C)$. Q is set via resistors R_2 and R_3 , where $R_3/R_2 = 3Q-1$. All resistors labeled R must be equal but may be set to any convenient value that will optimize other aspects of the circuit's performance (e.g. noise, etc.) because the value of R does not influence the F_P and Q of the circuit. Choosing $R_3 = R$ leaves R_2 to set the Q. The two integrators formed by A2 and A3 do not require high precision value capacitors because the values of R_{F1} and R_{F2} can be trimmed to compensate for capacitance variation from the desired values as long as the RC product for each integrator remains the same, e.g. $RC = 1/(2\pi F_P)$.

Configuring the circuit of figure 2 for the desired notch response and unity passband gain is carried out via the following steps: First, the values of C_{F1} , C_{F2} , R_{F1} , and R_{F2} are selected to produce the desired values of F_P and Q following the procedure outlined above. To form a lowpass notch, the summing amplifier gain applied to the lowpass filter output (LP) is set equal to 1 by setting $R_{LP} = R_1$. This sets the desired passband level of 0dB. Also, the summing amplifier gain for the highpass filter output (HP) is set to the value $(F_P/F_Z)^2$ noting that $F_P < F_Z$. Since R_1 must remain the same, we set $R_{HP} = R_1 * (F_Z/F_P)^2$. The notch appears in the output (NOTCH) at the frequency where the level of the lowpass filter has fallen to the level of the scaled highpass filter due to the opposite phase of these two signals and as frequency is increased the gain rebounds and levels out at the value $(F_P/F_Z)^2$. In a similar manner a highpass notch can be generated by setting $R_{HP} = R_1$ and $R_{LP} = R_1 * (F_P/F_Z)^2$, noting that in this case $F_P > F_Z$. Any ratio of notch to pole frequency can be produced and easily adjusted via the resistors connecting the LP and HP outputs to the inverting summer. These qualities make this topology very versatile and higher in performance than other analog active notch filter circuits. Keep in mind that for an Nth order elliptic filter up to N/2 state variable filters will be required since each is of order 2.

IIR digital filters can also be used to implement all the notch responses. When using this realization, the lowpass notch gain correction is easily incorporated directly into the IIR filter coefficients. Unfortunately, the lowpass and highpass notch forms are typically not found among the pre-programmed filter types of hardware DSP platforms. In some cases the user can upload filter coefficients directly, so a short C++ program for calculating them is provided in the document 'C++ code to generate notch transfer function coefficients'. There are a few software DSP applications that include lowpass and highpass notch filters. These include the author's ACDf LADSPA plugin [3] and the CamillaDSP application [4]. The ACDf notch filter implements equation 1 directly, so the user must explicitly provide the gain correction factor as part of each lowpass notch. For IIR filters, the gain correction must be calculated using pre-warped pole and zero frequencies, e.g. $40 \log_{10}(\tan(\pi F_P/F_S) / \tan(\pi F_Z/F_S))$ where F_S is the sampling rate. The author

is indebted to Dr. Ulrich Brüggemann for his advice regarding this formula. In the CamillaDSP application the “General Notch” IIR filter type should be used. A parameter of the “General Notch” filter called “normalize_at_dc” will automatically perform the gain correction so that the lowpass notch passband has gain of 0dB. This parameter is enabled by default.

IIR digital implementations have several advantages. The responses are mathematically exact and not affected by component tolerances or drift. The signal to noise ratio remains very high no matter how much processing is performed on the signal. Digital filters can easily be reprogrammed when a change in the response is desired, and this is a rapid process. On the downside there will be some amount of latency due to the overhead of processing digital data. Latency is greatest for computer-based DSP programs, but also exists to a much smaller extent on hardware-based DSPs. Only the analog active filter implementation is truly “real time”.

In addition to filter implementation, it is useful to be able to design an elliptic filter such that it is expressed in terms of first and second order stages. There is a method for elliptic filter design by Dimopoulos [5] that is highly recommended (see chapter 4). Using Dimopoulos’ notation, the user supplies the desired normalized stopband cutoff frequency $\Omega_s = F_{stop}/F_c$, ripple amplitude A_{MAX} in dB, and stopband attenuation A_{MIN} in dB. The algorithm produces the required filter order and the values of Ω_p , Ω_z , and Q_p for each second order stage, as well as Ω_p of any real poles (first order stages). I have provided an MS-Excel design aid titled ‘Elliptic Filter Design from Specifications’ that performs these calculations so that the user can simply plug in the desired parameters to obtain the necessary normalized frequency values. Ω_p and Ω_z are then multiplied by the desired corner frequency F_c to obtain F_p , F_z and along with Q_p can be implemented via the SVF analog circuit shown in figure 2 or using IIR digital filtering. This should simplify for the reader the process of designing and implementing elliptic or similar filters.

The reader should now feel more familiar with the lowpass and highpass notch responses, their implementation, and the design of elliptic filters using the methods and aids described above. These filters can play useful roles when sharp HP or LP band-limiting of audio signals is needed. Other types of filters with notches can be specified and implemented in the same manner and experimentation in this area may lead to new filters for audio applications, provided that the shoe fits, of course.

REFERENCES:

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